Experimental Demonstration of a Low-complexity Multiple-input Single-output Frequency Diverse Array Framework

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***Abstract*—Spatially diverse transmissions provide spatial selectivity via a fast-time modulation structure that changes with angle. This property provides the ability to spatially beamform on receive via signal separation according to the well-known virtual array. Therefore, a single channel receiver can be used in a Multiple-Input Single-Output (MISO) configuration to provide delay, Doppler, and angular information of scatterers illuminated with this type of transmission. Here, we experimentally demonstrate a MISO concept with a linear frequency modulated continuous-wave frequency diverse array (LFMCW-FDA) framework. This transmission is investigated for its ease of implementation, its applicability to standard receive signal processing, and benefits regarding the usable unambiguous range-Doppler space compared to other spatially diverse transmissions (i.e., Doppler-division and time-division). The experimental results show that, as predicted, the MISO virtual aperture for the LFMCW-FDA transmission appears in the delay dimension after correlation filtering and contains the spatial information of the scatterer.**

***Index Terms*—Multiple-input single-output radar, frequency diverse array**

I. Introduction

Persistent illumination of a radar scene is an attractive means of obtaining long dwell times without the timeline constraints of traditional scanning radars. The ubiquitous (or "staring") radar is an example of this type of system, where a broad transmit beam illuminates a large volume and multiple beams are formed on receive to detect and spatially discriminate scattering within the volume [1]. Persistent illumination can also be accomplished via a spatially diverse transmission, where multiple independent waveforms are transmitted such that the aggregate emission is a wide transmit beam with a spatially varying modulation structure providing spatial selectivity on receive. Spatially diverse transmissions are typically associated with collocated multiple-input multiple-output (MIMO) systems (i.e., multiple transmit and receive antennas) [2]–[4], and are typically referred to as "MIMO waveforms" even though they can be implemented in a single receiver context as well [5]. This configuration is called multiple-input single-output (MISO) which can be seen as the converse of the ubiquitous radar concept where the spatial degrees of freedom are pushed to the transmit array rather than the receive array.

MIMO waveforms are constructed via independent modulation structures that provide "separability" between the transmitted waveforms. For example, Doppler-division multiple access (DDMA) and time-division multiple access (TDMA) implement a spatially diverse transmission through either frequency or time shifting the transmitted waveforms relative to each other [2, 3]. The frequency diverse array (FDA) is another type of spatially diverse transmission where each antenna transmits a different frequency offset relative to a carrier [6]–[8]. In this paper, we investigate a linear frequency modulated continuous wave (LFMCW) version of FDA called LFMCW-FDA, which provides an interesting trade space between DDMA and TDMA transmissions (variants of the LFMCW-FDA have been previously published in [3], [9]–[11]).

To demonstrate the spatial selectivity of the transmission, LFMCW-FDA is implemented in a multiple-input single-output context. This configuration has the benefits of obtaining range, velocity, and spatial estimates of scatterers with only a single receive antenna which may be required in a size, weight, and power constrained (receiver) configuration. It is shown that the LFMCW-FDA processing can be implemented via a single correlation filtering operation followed by a beamforming stage (to obtain a range-angle estimate), and can optionally be extended to obtain velocity estimates when aggregating multiple transmitted sweeps. It is shown that, when compared to similar DDMA and TDMA implementations, the LFMCW-FDA transmission retains the full unambiguous range-Doppler space (DDMA and TDMA reduce the unambiguous space by a factor equal to the number of transmit elements) but has a loss in effective bandwidth (also observed in [10, 11]).

A MISO demonstration of LFMCW-FDA is provided which highlights the spatial estimation of a known mobile scatterer in the scene. It is shown that the spatial aperture of the LFMCW-FDA extends in the delay (range) dimension due to the delay-Doppler coupling of the LFM ambiguity function. This "virtual

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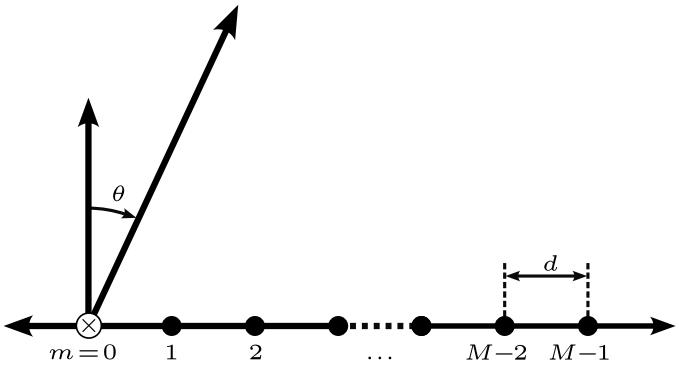

Fig. 1. M Element Uniform Linear Array (ULA) geometry.

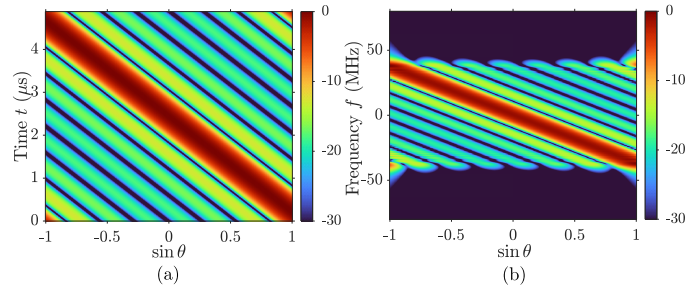

Fig. 2. Far-field emission structure for a single sweep: (a) Time-varying beampattern $|g(t,\theta)|^2$ (dB), (b) Far-field energy spectral density $|G(f,\theta)|^2$ in (dB).

array" is shown experimentally by observing the angle of the known scatterer after range-Doppler processing but before spatial compression.

II. Transmit Signal Modeling

The transmit array is designed to emit a frequency-shifted LFMCW sequence from each element. This transmission does not require independent and arbitrary control over the waveforms transmitted by each element (i.e., a digital array) and can be implemented in a low complexity framework via a frequency diverse array concept [6] (a motivating factor for constraining the transmission to this form). Consider an M element narrowband uniform linear array (ULA) with inter-element spacing d on transmit, as shown in Figure 1. Assuming an up-chirped LFMCW with progressively increasing frequency across the array, the complex-baseband waveform emitted by the m-th element is defined as

$$s_m(t) = \sum_{n=0}^{N-1} s(t - nT) e^{j2\pi m \Delta F t}, \tag{1}$$

where N is the number of chirps in the sequence, T is the chirp duration, ΔF is the frequency offset between channels, and

$$s(t) = \begin{cases} e^{j2\pi(-\frac{B}{2}t + \frac{B}{2T}t^2)} & 0 \le t < T \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

is a single baseband LFM of bandwidth B.

After applying an initial steering phase to each waveform, the complex envelope of the far-field emission as a function of azimuth angle θ (assuming omnidirectional antenna patterns) can be found by applying the ULA steering vectors as

$$g(t,\theta) = \sum_{m=0}^{M-1} \left(s_m(t) e^{jm\phi_0} \right) e^{j\frac{2\pi F_c}{c} m d \sin\theta}, \tag{3}$$

where ϕ_0 is the initial steering phase, c is the speed of light in free space, and F_c is the center frequency.

The LFM and FDA repetition intervals (T for LFM and $\frac{1}{\Delta F}$ for FDA) are in general decoupled; however, for this work we enforce $\Delta F = \frac{1}{T}$ to ensure equal duration for spatial scan and frequency sweep. This approach has the benefit of relating swept frequency to scanned angle and produces desirable range sidelobe performance (see [3, 9] for ambiguity diagrams of similar transmissions). The term "sweep" will refer to a single instance of the repetition interval T (single spatial scan and frequency sweep). Furthermore, the initial steering direction ϕ_0 is chosen such that the mainbeam of the transmission is at boresight midway through a sweep ($t = \frac{T}{2}$). This occurs when the summation of the element-dependent phasing at this instant is equal to zero,

$$0 = \phi_0 + 2\pi \Delta F t \Big|_{\Delta F = \frac{1}{T}, t = \frac{T}{2}}, \tag{4}$$

which results in an initial steering phase of $\phi_0 = -\pi$.

An example of the time-varying beampattern $|g(t,\theta)|^2$ for a single sweep is shown in Figure 2a for the case of an $M = 8$ element ULA with $d = 4.13$ cm, $B = 81.92$ MHz, $F_c = 3.3$ GHz and $\Delta F = 204.8$ kHz (recall the sweep duration T is the inverse of the frequency offset ΔF). These parameters are the same used in the experimental demonstration discussion in Section V. The energy of the LFM transmission is swept continuously over all $\sin\theta$ space in a uniform manner.

The magnitude-squared of the Fourier transform of $g(t,\theta)$ over time yields the far-field energy spectral density $|G(f,\theta)|^2$. The energy spectral density of the transmission in Figure 2a is shown in Figure 2b. Enforcement of $\Delta F = \frac{1}{T}$ is demonstrated by the swept pattern over space for both time and frequency. The angular-dependent signal structure of the transmission enables spatial selectivity and estimation in the single channel receiver.

III. Receiver Development

To demonstrate the appearance of the virtual array and form the basis of the receive algorithm, consider a single scatterer with complex scattering coefficient γ at some delay τ_s, angle θ_s, and Doppler shift $F_{D,s}$. The scatterer signal response at a single channel receiver for the entire CPI is a scaled, delay and Doppler shifted copy of the far-field complex envelope in the direction θ_s (assuming no range-walking effects from envelope dilation),

$$y(t) = \gamma g(t - \tau_s, \theta_s) e^{j2\pi F_{D,s}(t - \tau_s)}. \tag{5}$$

Consider the signal response received for the initial sweep of the LFMCW sequence ($n = 0$) with zero Doppler frequency

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$F_{\mathrm{D,s}} = 0$ Hz. Defining this response and inserting the form of $g(t,\theta)$ yields

$$\tilde{y}(t) = \gamma \sum_{m=0}^{M-1} s(t-\tau_{\mathrm{s}}) e^{j2\pi \frac{m}{T}(t-\tau_{\mathrm{s}})} e^{-jm\pi} e^{j\frac{2\pi F_{\mathrm{c}}}{c} md \sin\theta_{\mathrm{s}}}. \quad (6)$$

Because we assume an LFM waveform for $s(t)$, the delay-Doppler coupling of its ambiguity response can be leveraged to require only a single compression filter on receive. The received signal $\tilde{y}(t)$ is correlated with a representation of the LFM signal *unshifted in frequency* to obtain an estimate of the scattering in delay,

$$\begin{aligned} x(\tau) &= \frac{1}{T}\int_{-\infty}^{\infty} s^*(t-\tau)\tilde{y}(t)dt \\ &= \gamma \sum_{m=0}^{M-1} A\left(\tau-\tau_{\mathrm{s}}, \tfrac{m}{T}\right) e^{-jm\pi} e^{j\frac{2\pi F_{\mathrm{c}}}{c} md \sin\theta_{\mathrm{s}}}, \end{aligned} \quad (7)$$

where

$$A(\tau, F_{\mathrm{D}}) = \frac{1}{T}\int_0^T s^*(t-\tau)s(t)e^{j2\pi F_{\mathrm{D}} t}dt \quad (8)$$

is the delay-Doppler ambiguity function of $s(t)$. The LFM ambiguity function within the summation can be approximated as delay and phase shifted copies of the autocorrelation,

$$A(\tau, \tfrac{m}{T}) \approx a\left(\tau + \tfrac{m}{B}\right) e^{-j\pi \frac{1}{BT} m^2} e^{jm\pi}, \quad (9)$$

where $a(\tau) = A(\tau, 0)$ is the autocorrelation function[1]. The phase term that is a quadratic function of m^2 is a residual video phase term[2] that is undesirable; however, this term can either be considered negligible (if BT is "large") or can be compensated for during spatial processing. The video phase correction will be shown when discussing spatial compression. The resulting delay response of the scatterer for a single sweep after matched filtering to an unshifted LFM can be approximated as

$$x(\tau) \approx \gamma \sum_{m=0}^{M-1} a(\tau - \tau_s + \tfrac{m}{B}) e^{-j\pi \frac{1}{BT} m^2} e^{j\frac{2\pi F_{\mathrm{c}}}{c} md \sin\theta_{\mathrm{s}}}. \quad (10)$$

Figure 3 shows an example of a delay compressed response from a single receive antenna receiving backscatter from a target at boresight ($\theta_{\mathrm{s}} = 0°$) after cross-correlating with an unshifted LFM $s(t)$. The transmitter is an $M = 8$ element array transmitting LFM waveforms of time-bandwidth product of $BT = 100$ and frequency offsets of $\Delta f = 1/T$. The top plot shows each individual response of the cross-correlation representing each transmit element, and the bottom plot shows their superposition. The phasing encoded onto this "spatial" aperture corresponds to the angle from which it scattered;

[1]The reason we can make this approximation is due to two factors: 1) the energy loss due to reduced overlap between envelopes is assumed negligible, and 2) the assumption that the distortion to the correlation function caused by a (small) reduction in bandwidth is negligible.

[2]The term "video phase" comes from a residual phase term that arises from a stretch processed LFM. We adopt this terminology due to the similarity in form and the fact that it is undesired.

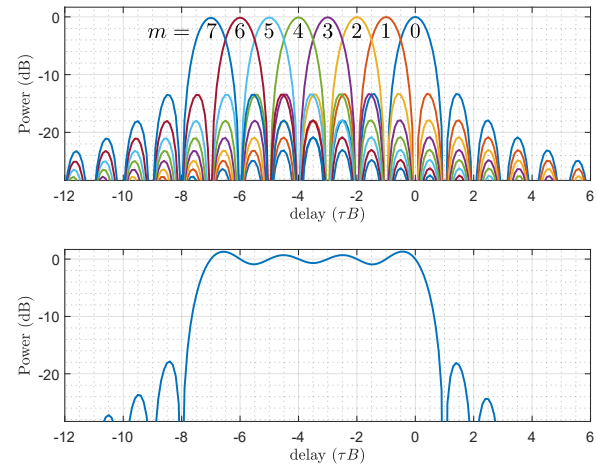

Fig. 3. Individual (top) and superposition (bottom) delay compressed responses for an $M = 8$ element ULA for a theoretical scatterer at boresight ($\theta_{\mathrm{s}} = 0°$). Response forms the virtual array on which spatial processing can be performed.

therefore, we can perform spatial processing using only a single receive antenna (likewise shown in [5]).

Note that the collection of autocorrelation functions in (10) are shifted by $+\frac{m}{B}$, thus the response of higher values of m will appear closer in delay. In Figure 3, the antenna element index m corresponding to the individual responses decrease from left to right (lower delay to higher delay). This sequencing is a result of implementing an up-chirped LFM paired with an increasing frequency offset $\Delta f = \frac{1}{T}$. We can reverse this sequencing by selecting either a down-chirp or decreasing frequency offset $\Delta f = -\frac{1}{T}$; however, if both are selected we obtain the same reversed element sequencing as for an up-chirp with positive frequency offset.

The virtual array of the scatterer can be beamformed on receive by replicating the received signal M times and delay shifting each copy by multiples of the delay resolution $\frac{k}{B}$. This process will align the M correlation responses of a scatterer across the virtual elements. The replicated and delay shifted copies of $x(t)$ containing the scattering can then be corrected according to the expected video phase rotation and beamformed to a desired azimuth angle as

$$x(\tau,\theta) = \frac{1}{M}\sum_{k=0}^{M-1} x(\tau - \tfrac{k}{B}) e^{j\pi \frac{1}{BT} k^2} e^{-j\frac{2\pi F_{\mathrm{c}}}{c} kd \sin\theta}. \quad (11)$$

This algorithmic process allows us to develop a delay-angle ambiguity function for the expected response of this receiver as a function of scatterer angle θ_{s}, beamformed angle θ, and delay τ. Using the forms of (7) and (11) and assuming a target delay of $\tau_{\mathrm{s}} = 0$ and unit amplitude $\gamma = 1$, the delay-angle ambiguity function can be written in terms of the LFM ambiguity function $A(\tau, F_{\mathrm{D}})$ as

$$\begin{aligned} &\bar{A}(\tau,\theta,\theta_{\mathrm{s}}) \\ &= \frac{1}{M}\sum_{k=0}^{M-1}\sum_{m=0}^{M-1} A\left(\tau - \tfrac{k}{B}, \tfrac{m}{T}\right) e^{j\frac{2\pi F_{\mathrm{c}}}{c} md(\sin\theta_{\mathrm{s}} - \sin\theta)} e^{-jm\pi} e^{j\frac{\pi k^2}{BT}}. \end{aligned} \quad (12)$$

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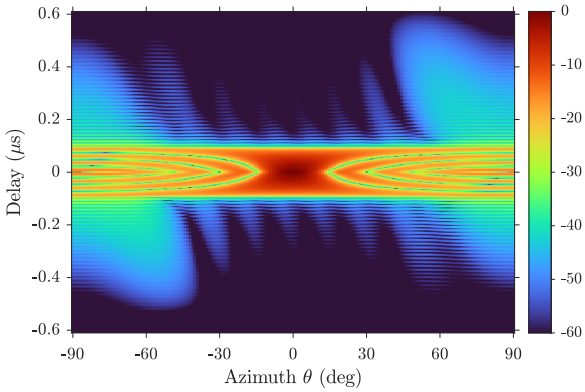

Fig. 4. LFMCW-FDA delay-angle ambiguity function $\bar{A}(\tau, \theta, \theta_s)$ for $\theta_s = 0°$ for a bandwidth $B = 81.92$ MHz, pulsewidth $T = 4.8828$ μs, and $M = 8$ element ULA with $d = \frac{c}{2F_c}$.

Figure 4 shows the expected delay-angle response of a scatterer at boresight $(\theta_s = 0°)$ for an LFMCW-FDA transmission with a bandwidth of $B = 81.92$ MHz, pulsewidth $T = 4.8828$ μs, frequency offset $\Delta F = \frac{1}{T} = 204.8$ MHz from an $M = 8$ element ULA with $d = \frac{c}{2F_c}$. This transmission structure creates a natural tapering effect in time/frequency according to the array beampattern, which (for a ULA) produces a sharp decrease in delay sidelobe level outside of the mainlobe region [3, 9].

While a continuous-time representation of the receive signal is beneficial from a modeling and intuition perspective, the received signal is inevitably sampled by an analog-to-digital converter. Represent the discretized delay-compressed complex-valued signal from (7) as $\mathbf{x}$, which we assume is critically sampled, $F_s = B$, where F_s is the sampling rate. Sampling the data at this rate produces a compressed delay response that is sampled at delay resolution of the chirp $1/B$; therefore, the virtual spatial array will consist of M consecutive samples over which we can perform spatial estimation. Defining the M pulse compressed samples corresponding to sample delay ℓ as $\mathbf{x}_\ell$, the scattering estimate at the ℓth delay sample and angle θ can be expressed as

$$\hat{x}(\ell, \theta) = \frac{1}{M}\mathbf{v}^H(\theta)\mathbf{J}\mathbf{x}_\ell, \tag{13}$$

where

$$[\mathbf{v}(\theta)]_m = e^{-j\pi\frac{1}{BT}m^2} e^{j\frac{2\pi F_c}{c}md\sin\theta} \tag{14}$$

is the mth sample of the spatial steering vector for azimuth angle θ with the video phase correction, and $\mathbf{J}$ is a rotation matrix that flips the indices of $\mathbf{x}_\ell$. The matrix $\mathbf{J}$ can be removed if the delay aperture (as shown in Figure 3) is not a reversed element sequencing. Furthermore, Doppler estimation can be implemented by collecting the delay estimates from multiple sweeps in slow-time and performing Doppler compression to obtain scattering estimates in delay, Doppler, and space. All processing steps described here can be implemented with low complexity operations.

IV. Comparison to other spatially diverse transmission methods

Taking an angular slice of the energy spectral density in Figure 2b shows that the effective bandwidth of the signal (in the given direction) is reduced by a factor approximately equal to the number of elements compared to the full LFM bandwidth. This reduction in bandwidth (and commensurate spoiling in delay resolution) is a result of the chosen frequency spacing $\Delta F = 1/T$. The reduction in bandwidth is similar to reductions in dimensionality for other common MIMO transmission methods (e.g., DDMA and TDMA). For DDMA transmissions, the Doppler dimension is apportioned for each transmitted waveform through application of a progressive slow-time phasing such that the backscatter from each Doppler-shifted waveform is orthogonal in the Doppler dimension [12]. For DDMA to be orthogonal, the Doppler bandwidth of the backscatter must be less than PRF/M, effectively reducing the usable unambiguous range-Doppler space by a factor of M. Likewise, a TDMA waveform partitions the delay dimension to orthogonalize the backscatter from each transmitted waveform. For TDMA to be orthogonal, the delay spread of the backscatter must be less than PRI/M reducing the usable unambiguous range-Doppler space by a factor of M.

The method discussed here (and in previous publications [3, 10]) frequency-shifts the transmitted waveforms much like DDMA; however, a frequency shift of $\Delta F = 1/T$ ensures that the waveforms are orthogonal at zero delay correlation[3] for a single sweep (assuming a constant amplitude waveform). Furthermore, as shown in (7) and (10), by assuming an LFM modulation structure for the transmitted waveform, the delay-Doppler coupling of the LFM ambiguity diagram can be leveraged to only require a single correlation operation on receive. For $\Delta F = 1/T$, the responses from each LFM show up delayed by $\tau_m = m/B$ (multiples of the delay resolution); therefore, the individual responses of each transmit path are orthogonal and a virtual array can be formed for subsequent spatial processing.

Unlike DDMA and TDMA, the LFMCW-FDA structure maintains the maximum usable range-Doppler space by arranging the virtual array in the delay dimension separated by the delay resolution of the LFM. However, the trade-off lies in the subsequent reduction in effective bandwidth seen in any particular direction. The virtual array in this context effectively becomes the new delay (range) resolution of the system. Therefore, this type of transmission is advantageous for cases when the unambiguous range-Doppler space cannot be partitioned without interference between virtual channels (e.g., high Doppler and delay spread of backscatter), and when high bandwidth is not a requirement (e.g., in a low-resolution search mode).

[3]The cross-correlation between waveforms will not be zero at non-zero delay.

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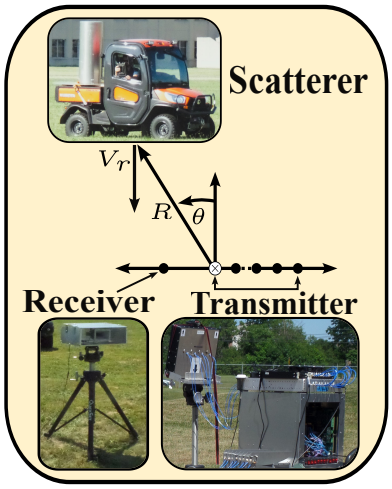

Fig. 5. Experiment scene layout

TABLE I
SUMMARY OF OPERATING PARAMETERS

Description	Value
Center Frequency F_c	3.3 GHz
LFM Bandwidth B	81.92 MHz
Sweep time T	4.8828 μs
Frequency offset ΔF	204.8 kHz
CPI Duration	10 ms
Number of Sweeps	2048
ADC rate	245.76 MHz
DSP sampling rate F_s	81.92 MHz
Transmit Channels M	8
Receive Channels	1

V. EXPERIMENTATION

To experimentally verify the LFMCW-FDA theory, a MISO LFMCW-FDA system configuration for moving target indication was tested at Wright-Patterson Air Force Base in an open-air environment. As shown in Figure 5, the experiment consisted of an $M = 8$ channel array on transmit, a known scatterer by means of a mobile "tophat" reflector strapped to an all-terrain vehicle, and a single channel horn receiver. The experimental results shown here consist of range, velocity, and spatial parameter estimates of the mobile tophat within a single coherent processing interval.

A. Experiment System and Parameters

A Zync UltraScale+ XCZU29DR RFSoC was used as the transceiver for the demonstration providing the necessary eight transmit channels and single receive channel. The RFSoC implemented eight individual LFMCW sequences of bandwidth $B = 81.92$ MHz with progressively increasing frequency offset of $\Delta F = 204.8$ kHz. As discussed previously, the sweep time is enforced to be $T = 1/\Delta F = 4.8828$ μs. The transmit waveforms were implemented within the RFSoC using digital up-conversion to center frequency of 3.3 GHz. An amplification stage was added to the transmitter chain to achieve a maximum desired output power of 30 dBm per channel, which were then attached to an S-band antenna array with a uniform horizontal spacing of 4.1275 cm. The single receive channel consisted of an S-band horn antenna connected to a low-noise amplification stage and sampled using the RFSoC, where the data was digitally down-converted to baseband and stored to memory at a rate of 245.76 MHz. The data was then post-processed at a resampled rate of 81.92 MHz. The transmit array and receive antenna were placed in a quasi-monostatic configuration for simultaneous transmit and receive. The operating parameters used for the collection are summarized in Table I.

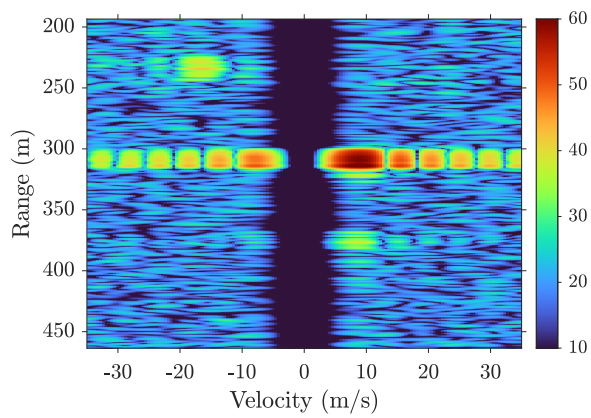

Fig. 6. Range-Doppler estimate (in dB) of scattered scene prior to spatial compression.

B. Experimental Results

The results shown in this section correspond to a 10 ms collection of the LFMCW-FDA transmission with parameters shown in Table I, which corresponds to 2048 total sweeps. For this collect, the mobile tophat was inbound towards the radar at negative azimuthal angle. The ground (stationary) clutter is suppressed via a low frequency clutter rejection filter.

Figure 6 shows the range-Doppler estimate of the scene (with ground clutter suppressed) prior to spatial compression (tantamount to Doppler processing multiple sweeps of the signal structure in (7)). A large scatterer corresponding to the mobile tophat can be clearly seen at approximately 314 m with an apparent velocity of approximately 9 m/s. The radial velocity is positive (i.e., towards the radar) as expected. Smaller "scatterers of opportunity" can also be seen at apparent ranges of 230 m and 360 m. These smaller responses can be attributed to traffic from a nearby road that is folding in range.

Figure 7 shows a zoomed-in look at the response of the mobile tophat from Figure 6 and up-sampled in range and Doppler for improved visualization (left image is magnitude in

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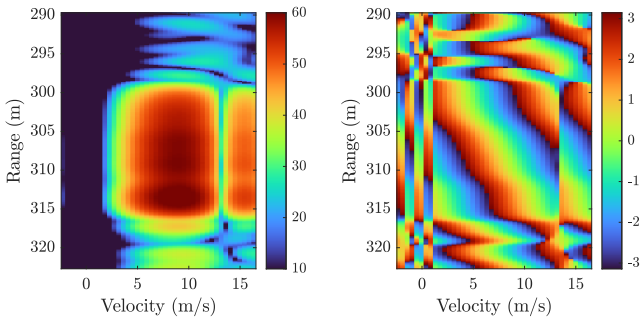

Fig. 7. Zoomed-in visual of the mobile tophat scattering response in range-Doppler. Left image shows the zoomed-in virtual array of the scatterer prior to spatial compression. Right image shows the phase response of the scatterer.

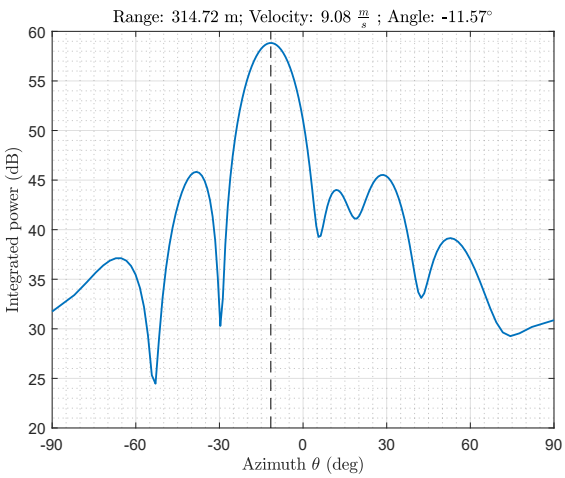

Fig. 8. Spatial cut at the range-Doppler location of the mobile tophat displaying the estimated parameters.

dB, right is phase in radians). Since this data is prior to spatial compression, the extended response in delay corresponds to the virtual array response from each transmitted waveform which will have the spatial information embedded in the phase of the response. The right plot of Figure 7 shows the phase response of the mobile tophat, which appears as a linear phase progression. Therefore, the angle of arrival can be determined via estimation of the slope of this phase progression accomplished via the beamforming operation described in (13).

The spatial selectivity of the virtual array can be observed by beamforming the virtual array samples corresponding to the scatterer. Figure 8 shows the spatially beamformed response as a function of azimuth angle θ for the delay samples corresponding to the virtual array of the mobile tophat. The beamformed response has the resolution of the transmitted array as predicted for a MISO configuration [5]. It is estimated from this response that the mobile tophat is inbound at an angle of $\theta = -11.57°$. Figure 9 shows the range-Doppler response after spatial compression to the scattering angle of arrival. In comparison to Figure 6, there is a noticeable increase in dynamic range and suppression of the smaller "scatterers of opportunity" indicating that they are arriving from different

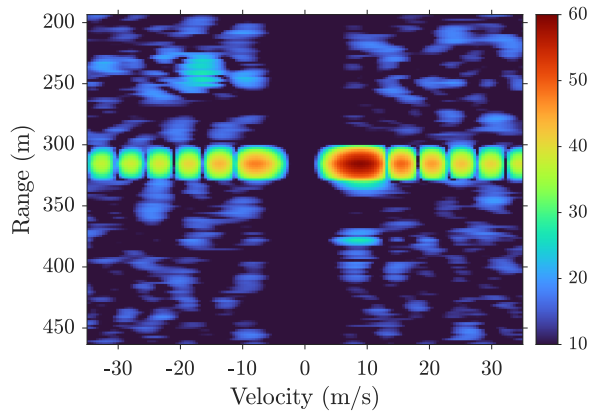

Fig. 9. Range-Doppler estimate (in dB) of scattered scene after spatial compression to $\theta = -11.57°$.

angles. While not included in these results, Doppler and/or spatial windowing could be employed to suppress unwanted sidelobes if desired.

VI. Conclusions

In this paper, we experimentally demonstrated the viability of a spatially diverse transmission structure called LFMCW-FDA in a MISO architecture. When compared to traditional MIMO waveforms (i.e., DDMA and TDMA), this structure has the benefit of retaining the unambiguous range-Doppler space albeit at a commensurate loss in range resolution. Therefore, this architecture is most applicable for size, weight, and power constrained receivers and scenarios where the loss in effective bandwidth is tolerable (e.g., in a low-resolution search mode).

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